
Unit – 2 : Session – 12 : SLO – 2 | Semaphore Types and Operations — Fill in the Blanks

Semaphores

1. A counting semaphore can have a value greater than 1 and is used for managing multiple resources.
2. A binary semaphore only takes the values 0 and 1 and is commonly used for mutual exclusion.
3. The wait operation decreases the semaphore value and may block the process if the value becomes negative.
4. The signal operation increases the semaphore value and wakes up waiting processes if any.
5. The traditional names for the wait and signal operations are P() and V() respectively.
6. Semaphores are widely used to achieve process synchronization and coordinate access to the critical section.
7. Improper use of semaphores can lead to deadlock, where processes wait indefinitely for resources.
8. A semaphore maintains an internal counter (and a queue of waiting processes) to keep track of its value and waiting processes.
9. Semaphores help in resource management by ensuring that only a limited number of processes can access shared resources simultaneously.
10. Using semaphores correctly helps avoid race conditions in concurrent programming.